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Paleomagnetism of granites from the Angara-Kan basement inlier, Siberian craton

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Abstract

We report a new paleomagnetic determination of Paleoproterozoic rocks from the Siberian craton which showed a positive baked contact test and a stable age of the high-temperature NRM component. The mean paleomagnetic pole of Siberia for ~1730 Ma located at 42.9° S, 109.6° E ($\alpha_{95} = 5.3°$) is compatible with the pole positions obtained recently for the middle and late Early Proterozoic.
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Keywords: Paleoproterozoic; paleomagnetic pole; Siberian craton

Introduction

Paleomagnetism has played a key role in the progress of geological thinking through the past fifty years. Among other hypotheses, it prompted periodic assembly and dispersal of supercontinents through the Earth's history, according to Wilson's cycles of ocean opening and closure (Khain, 2001).

The Late Paleozoic-Mesozoic supercontinent of Pangea has been reconstructed in more or less similar models, but the ideas of older Proterozoic supercontinents remain largely controversial. Such are the poorly consistent paleoreconstructions for the Siberian craton based mainly on structural and qualitative paleomagnetic data. The most recent overview of paleomagnetic knowledge on Precambrian cratons with implications for their configurations in the Proterozoic (Pesonen et al., 2003) logically contains only three poles for Siberia: from 1503 Ma dikes in the Kuonamka belt (Ernst et al., 2000), from the 1043 Ma Malga Formation (Gallet et al., 2000), and from 1005 Ma sills of the Kandy Formation (Pavlov et al., 2002).

A number of good paleomagnetic determinations obtained lately for Neoproterozoic Siberia (Metelkin et al., 2007; Pavlov et al., 2002) were in favor of a model (Rainbird et al., 1998; Yarmolyuk and Kovalenko, 2001) which placed it inside Rodinia assuming the connection between the Arctic margin of Laurentia and the Baikal margin of Siberia at least between 1070 and 950 Ma. High-quality paleomagnetic data for the Paleoproterozoic have become available only recently (Didenko et al., 2003, 2005, 2006, 2007; Vodovozov et al., 2007a,b) but they are obviously few to reconstruct the respective apparent polar wander path (APWP).

Thus, it appears urgent to obtain magnetically stable and reliably dated paleomagnetic directions for Middle and Early Proterozoic rocks from the Siberian craton, wherefrom to infer the Proterozoic APWP and to substantiate quantitatively the evolution of the Siberian plate and its possible location within the supercontinent. Some solutions for the problem may come from studying granites and related rocks. First, it is because granitoids are amenable to advanced isotope age methods and, second, their origin, in the existing genetic models, is associated with certain events in the tectonic history of cratons.

In 2007 we tried to derive new paleomagnetic directions from granitoids of the Taraka complex (sampling sites for paleomagnetism at 56.19° N, 94.93° E), which we assumed (Didenko et al., 2003) to be part of the Paleoproterozoic South Siberian postcollisional granite belt (Fig. 1, *A*), and from *A*-type hypersthene granites (locally called kuzeevites, after the Kuzeeva River), sampled at 56.77° N, 94.02° E in the Angara-Kan basement inlier in the southwestern Siberian craton (Fig. 1, *B*).

Laboratory measurements showed a very low paleomagnetic stability of the Taraka granitoids and, on the contrary, a high stability of the Angara-Kan granites. Thus below we report paleomagnetic data for the latter.

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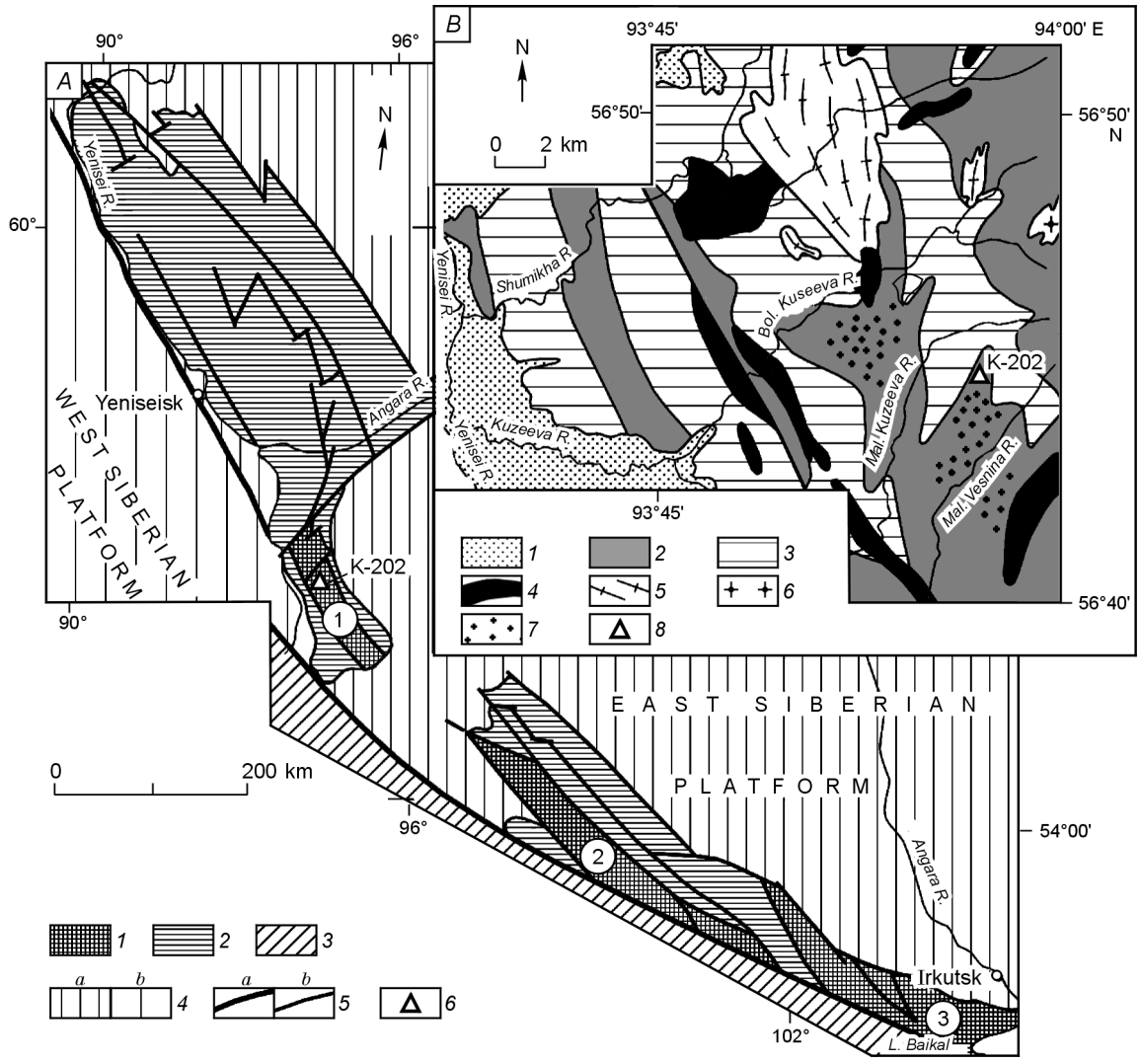


Fig. 1. Location map of Kan metamorphic complex (*A*) and granitoids (*B*), after (Bibikova et al., 2001; Nozhkin et al., 2003). *A*: *1* — basement of East Siberian platform; *2* — Late Proterozoic terrains; *3* — rocks of Sayan-Baikal folded area; *4* — Phanerozoic sedimentary cover of East Siberian (*a*) and West Siberian (*b*) platforms; *5* — main tectonic boundaries: margin of East Siberian platform (*a*), large faults (*b*); *6* — sampling sites. Numerals in circles are: Angara-Kan (1), Biryusa (2), and Sharyzhalgai (3) inliers. *B*: *1* — Quaternary sediments, *2*, *3* — Kan metamorphic complex: Kuzeeva Group of garnet-biotite, garnet-hypersthene-biotite gneiss, and migmatite (*2*) and Atamanovka Group of hypersthene-cordierite-sillimanite gneiss, amphibole-pyroxene and two-pyroxene schists, marble, migmatite (*3*); *4* — Kimbirka complex of metagabbro, gabbronorite, anorthosite; *5* — Posolnaya complex of gneiss granite; *6* — granite of Taraka complex; *7* — areas of granitoids; *8* — sampling site.

Geological background

Rocks of the Angara-Kan basement inlier, the Kan metamorphic complex (Kan Group), have been traditionally assigned to oldest Archean rocks of East Siberia (Kuznetsov, 1988), mostly proceeding from the presumed Archean age of granulite facies and charnockite. However, granulite metamorphism turned out to be younger according to the 1900±20 Ma U-Pb zircon ages of the Kan granulites and enderbites from migmatite veins (Bibikova et al., 1993).

The Taraka granitoids in the eastern Angara-Kan inlier occur as the Taraka pluton buried under Upper Precambrian sediments in the east and under Devonian sediments in the south. In the west the pluton shows intrusive or locally fault contacts with country-rock gneisses and migmatites. The central and eastern parts of the pluton are composed of 1837±3 Ma (Nozhkin et al., 2003) massive porphyritic to equigranular biotite granite and leucogranite of the main phase. The 3–4 km wide pluton's western periphery consists of biotite granodiorite and plagiogranite. The structural setting and chemistry of porphyritic granite provide reliable evidence of a postcollisional origin of the pluton (Nozhkin et al., 2003), i.e., it postdates the granulite metamorphism of ~1.9 Ga (Bibikova et al., 1993; Nozhkin, 1999).

Brownish or gray massive porphyritic hypersthene granite, granodiorite, and granosyenite (kuzeevites) (Kuznetsov, 1988) exist as small intrusions or thin dikes that bear no signature of later metamorphism, folding or shearing in granulite-facies *PT* conditions. The geometry of the intrusions is hard to constrain because of poor exposure. In the area we explored, granites most often occur among the Kuzeeva Group gneisses as apophyses cutting into the migmatite and metamorphic

banding and contain xenoliths of garnet-biotite and hypersthene gneisses. The dike-shaped granitic bodies, which crosscut synmetamorphic charnockite-migmatite (bogunite) rocks (Kuznetsov, 1988), have chilled margins, while traces of heating are found in the surrounding country rocks and in the xenoliths.

The 1734±4 Ma Angara-Kan granites (Bibikova et al., 2001) are obviously younger than the granulite facies completed at 1900±20 Ma, i.e., they were never involved into metamorphism and folding after their emplacement (at least in the inner part of the inlier).

Thus, we hypothesize that after the main metamorphic event had ended, the Angara-Kan inlier was fixed within the Siberian craton (was part of it) and experienced neither horizontal nor vertical rotation relative to the latter. Therefore, we may use the NRM directions we obtained for the granite samples in the geographic coordinates to constrain the position of the latest Paleoproterozoic paleomagnetic pole of Siberia.

Paleomagnetic directions were determined in 21 oriented samples collected from a single 2.5–3 m thick granite vein we discovered in a 20–30 km^2 area with numerous exposures of the Kuzeeva Group gneisses. The latter were sampled for the baked contact test, both at their contact with granites and away from it, where the rocks were out of the reach of heating from the thin intrusion.

Methods and results

The samples for paleomagnetism were collected by hand (with hammer and chisel) and oriented using a sun compass, with a site declination of +4.5° assumed in site mean directions. In the laboratory the oriented samples were cut with a diamond power saw into 2×2×2 cm cubes, two or three replica specimens from each piece of rock.

Thermal magnetization analysis was applied using a Curie balance to identify the magnetization carriers, and the magnetic fraction was found out to consist of almost pure or cation-deficient magnetite, with the 575 to 610 °C Curie temperatures. The temperature dependences of saturation magnetization of the granite samples showed prominent Q-type behavior indicating a high percentage of the ferrimagnetic fraction.

All specimens were subject to stepwise thermal demagnetization in a shielded ASC TD-48 furnace (USA); NRM were measured after each cleaning step on a JR-4 spinner magnetometer (Czech Republic). The demagnetization data were plotted in Zijderveld diagrams (Zijderveld, 1967); then statistical analysis was applied to NRM components (Fig. 2, *A–F*) and the site mean directions (Fig. 2, *G*) were calculated using codes from (Enkin, 1994).

The diagrams of almost all granite samples (18 of 21) were easy to interpret, with a high-temperature NRM component identifiable from 4–5 or more points (Fig. 2, *A–D*). Unit vectors in the stereoplots clustered around the mean of *Dec* = 348.4°, *Inc* = 19.5°, *K* = 25.1, $\alpha_{95} = 7.0°$ (Fig. 2, *G*). This direction is reported in geographic coordinates assuming that the Angara-Kan inlier never experienced rotation about the horizontal axis.

The diagrams of the gneisses were likewise well defined (Fig. 2, *E*, *F*). Note that the high-temperature NRM component of the gneiss samples showed a direction different from that of the granites (Fig. 2, *G*). Partial remagnetization at the time of emplacement can be expected only in the gneisses that immediately surround the intrusion (Fig. 2, *E*) whereas those away from the margin appear free from thermal effects (Fig. 2, *F*) and have a mean direction of *Dec* = 314.3°, *Inc* = –1.8°, *K*= 328, $\alpha_{95} = 5.2°$ (Fig. 2, *G*). The difference between the mean NRM directions of the granite and gneiss samples (without two samples from near the margin) is 39.7° (Fig. 2, *G*), or above the threshold of 30.5° corresponding to the *F* criterion ($\rho = 0.05$) for equality (possibility to belong to the same population) (Shipunov, 2000). Therefore, the two statistic samples belong to different populations.

Discussion and conclusions

It is impossible to infer from the available data the time when the gneiss samples acquired the high-temperature NRM component. This may be due to granulite-facies metamorphism and migmatization of the Kuzeeva rocks between 1840 and 1900 Ma (Bibikova et al., 1993; Nozhkin, 1999), i.e., ~100 Ma before the plutonic event. Proceeding from the statistical analysis of paleomagnetic data and the isotope ages of the intruded and country rocks, we infer that (i) there were no total remagnetization, and (ii) the NRM direction of granites corresponds to the time of intrusion, which is consistent with observation in the field. As examined in the field, the country rocks along the intrusion margin bear prominent signature of heating, i.e., granitic magma intruded into an already cold gneiss. Furthermore, the intrusion being quite small, its cooling could not be long and, hence, the high-temperature NRM record (blocking temperatures of 450–580 °C) could follow rather soon after crystallization of zircons (the U-Pb system in zircons closed at ~750–850 °C).

There is another argument for the latest Paleoproterozoic age of the paleomagnetic event. Recent paleomagnetic data for Paleoproterozoic rocks from the Siberian craton listed in Table 1 include several determinations with high stability, or the key poles, especially those for volcanic-sedimentary rocks of the Akitkan Group in the Baikal and Akitkan ranges (Table 1). The poles lie in a regular series from highly to less stable, and their positions on the sphere agree with the age succession, where the pole derived from the Angara-Kan granites fits quite well (Fig. 3).

The polarity problem is beyond the scope of this study, but we would mention that should the obtained poles (Table 1, Fig. 3) be the north poles (normal polarity), the Baikal margin of Siberia would juxtapose the Arctic margin of Laurentia inside the hypothetical supercontinent of Columbia, in a way as the available paleomagnetic data suggest for the latest Middle-earliest Late Proterozoic (Metelkin et al., 2007; Pavlov et al., 2002).

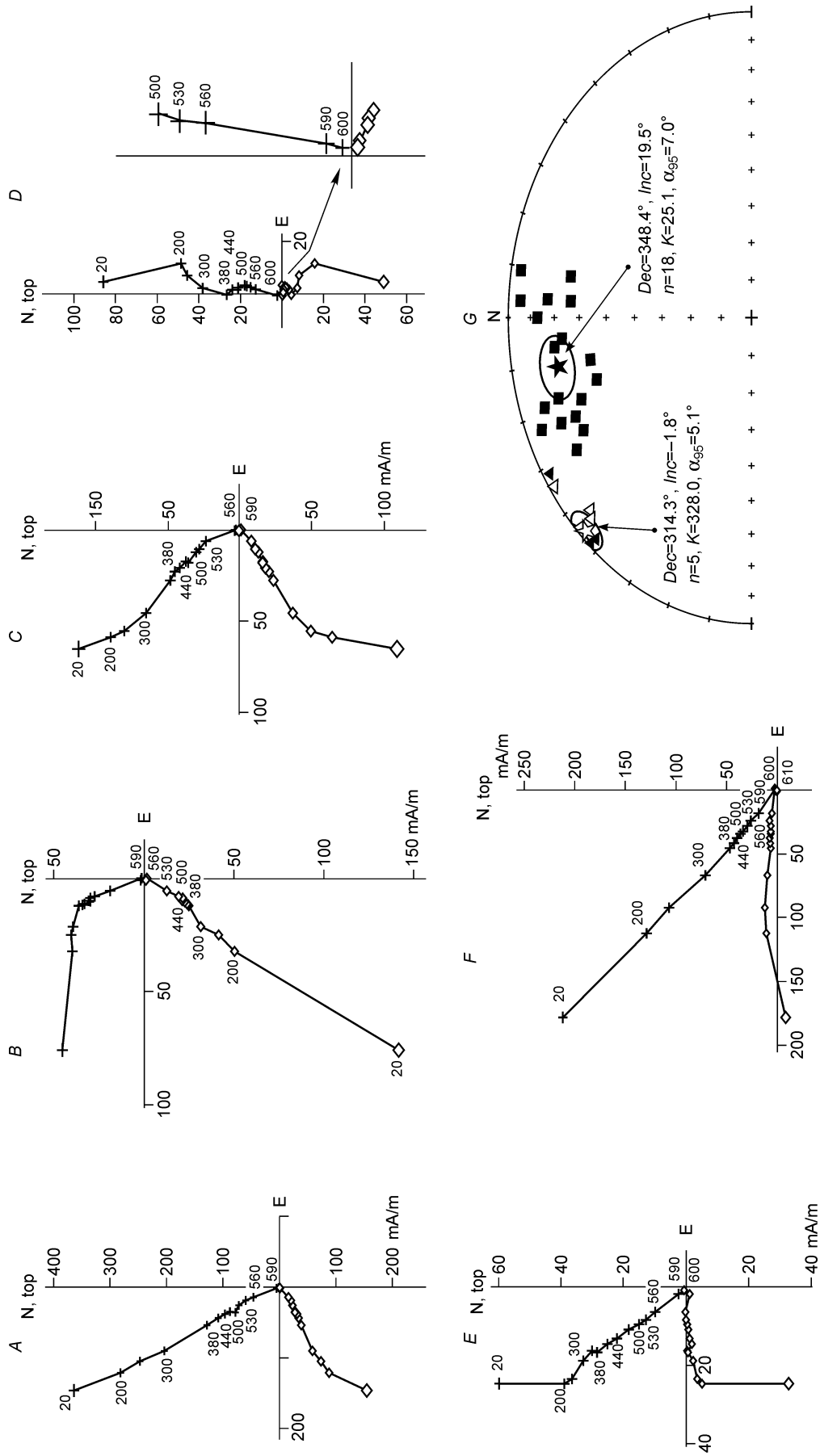


Fig. 2. Paleomagnetism of collected samples. *A–F*: thermal demagnetization data in Zijderveld diagrams. *A–D* — hypersthene granite samples D07-4/1, D07-4/4, D07-4/6, and D07-4/11, respectively; *E*, *F* — gneiss samples from Kuzeeva Group: D07-5/1 collected near intrusion (*E*) and D07-5/5 away from intrusion (*F*). Crosses and diamonds are, respectively, horizontal- and vertical-plane projections of NRM directions. Numerals at data points are temperatures in °C. *G*: high-temperature NRM directions measured in samples of granite (squares) and gneiss (triangles) in stereoplots (equal-area projection), in geographic coordinates. Asterisks are statistic mean paleomagnetic directions, with ovals of 95% confidence, in lower (bold symbol) and upper (open symbol) hemisphere projections. *Dec* is paleomagnetic declination, *Inc* is paleomagnetic inclination; *K* is kurtosis; α_{95} is radius of 95% confidence oval.

Table 1
New paleomagnetic directions and poles of Siberia

Pole, formation	Age, Ma	*Dec*, °	*Inc*, °	*K*	α_{95}, °	Test	Plat, °	Plong, °	*dp/dm*, °	Reference
Angara-Kan inlier										
1. Hypersthene granite	1734 ± 5	348.4	19.5	25.1	7.0	*C*	–42.9	109.6	3.8/7.3	This study, age after (Bibikova et al., 2001)
Sharyzhalgai inlier										
2. Shumikha granite	1854 ± 4	174.0	–26.0	9.4	5.9	*R*	–23.9	110.0	3.4/6.4	(Didenko et al., 2003, 2005, 2006)
North Baikal Range										
3. Malaya Kosa Formation	1877 ± 4	187.2	6.8	79.4	6.3	*G*, *F*	–31.7	100.2	3.2/6.3	(Didenko et al., 2007)
4. Dolerite dike	1674 ± 12	189.4	–16.2	23.5	4.0	*F*	–43.6	100.5	3.2/6.3	Complemented after (Vodovozov et al., 2007a); age after (Gladkochub et al., 2007)
Akitkan Range										
5. Khibelen Formation	1855 ± 4	207.4	16.1	23.5	7.3	*G*, *F*	–20.5	81.0	3.9/7.5	(Vodovozov et al., 2007b)
6. Chaya Formation	1863 ± 9	192.5	18.6	50.5	6.9	*C*, *G*	–22.1	97.5	3.7/7.2	(Didenko et al., 2006)
7. Okun' Formation	PR1	358.6	–8.1	14.7	9.6	*F*	–28.5	111.6	4.9/9.7	(Vodovozov et al., 2007b)
8. Gabbro dolerite	1674 (?)	193.3	–16.4	33.1	5.1	—	–39.6	92.8	2.7/5.3	Complemented after (Vodovozov et al., 2007a,b)

Note. Plat and Plong are latitude and longitude of paleomagnetic pole, respectively; *dp*, *dm* are semiaxes of 95% confidence oval. Letters stand for reversals (*R*), conglomerate (*G*), fold (*F*), and baked contact (*C*) tests. Pole numbers (1 to 8) correspond to those in Fig. 3. Other symbols as in Fig. 2.

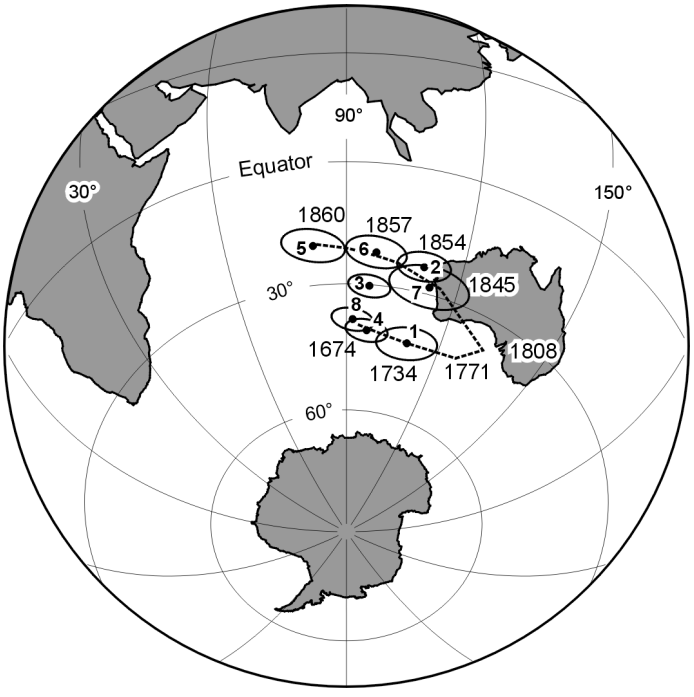


Fig. 3. Suggested APWP segment of Siberia for latest Paleoproterozoic. Points mark locations of new paleomagnetic poles with ovals of 95% confidence (pole numbers 1 to 8 correspond to those in Table 1). Dashed line is cubic-spline approximation (Torsvik and Smethurst, 1999) of APWP for Siberia. Numerals are ages in Ma.

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